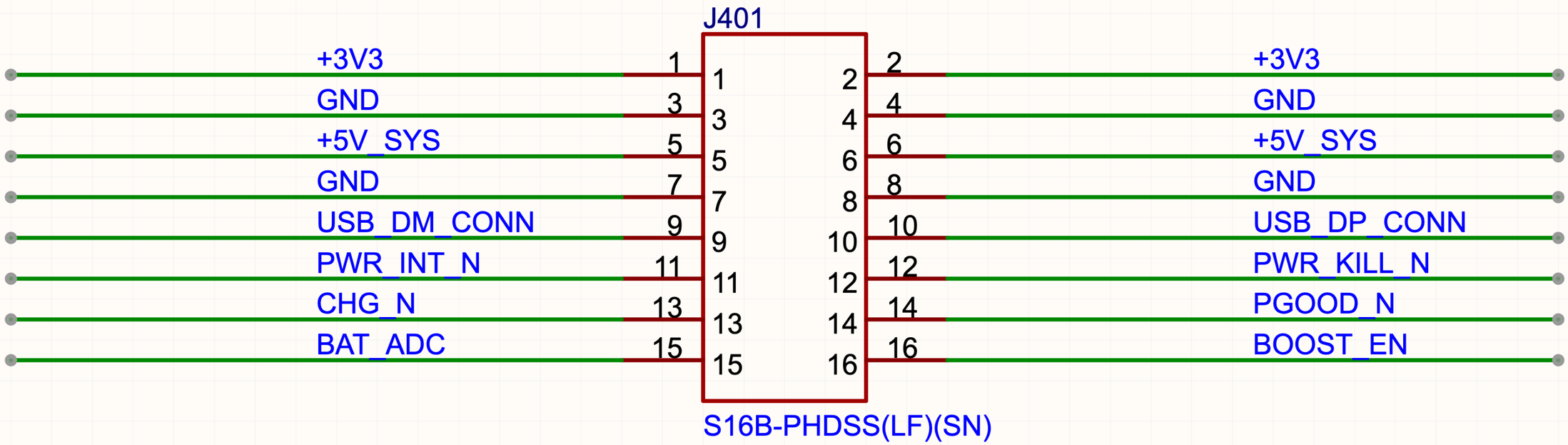
# V080 主控板 <-> 电源板 / 16-pin power + USB + control

J401 JST S16B-PHDSS(LF)(SN) / C161675 / 2.0mm 2x8




线束 H1：PHDR-16VS 两端，按针号 1->1 ... 16->16；禁止镜像接线。

1/2=3V3；5/6=5V；3/4/7/8=GND。电源/地用 AWG24，信号用 AWG28。

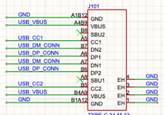
9/10 USB D-/D+ 用 90Ω 差分双绞线；首版线束长度目标 <=100mm。

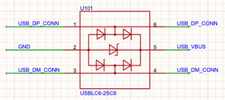
只在断电时插拔；线束固定并远离 ESP32 天线区域；本版本仅布局，未布线。

**SPIRIT / 10_USB / 原理图复核版 2026-10-18**

V074K: F101 BSMD1812-260-16V; BQ24075 selects USB500 or external ILIM from Type-C advertised current. F101 is not a 500mA current limiter.

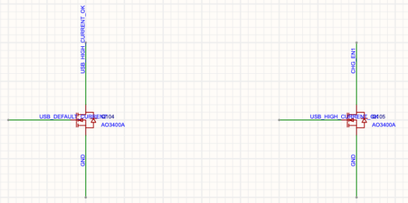
Default Rp / CC unpowered: USB500. 1.5A/3A Rp: ILIM R110=1.18k, approx 1.1-1.47A range including IC tolerance. Not USB-PD; USB enumeration/suspend compliance still requires validation.






















U111 TUSB320LAI replaces external Rd R137/R138. Q104/Q105 are newly assigned CC current logic, not the removed USB-presence detector. PGOOD_N remains GPIO39.

**SPIRIT / 11_Charger / 原理图复核版 2026-10-18**

CE through R108 low = charge enabled. EN2=USB_HIGH_CURRENT_OK; EN1 is its inverted signal pulled to VSYS_RAW. BQ24075 retains USB/battery power path.

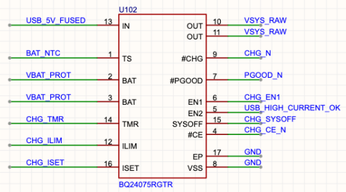
R111=1k sets nominal 890mA charging; real charge current is reduced by input limit, system demand, DPPM and temperature. It is not an extra load allowance.

R110=1.18k acts in ILIM mode only. CC default/unpowered -> EN1=1 EN2=0 USB500; Rp1.5A/3A -> EN1=0 EN2=1 ILIM. Main off -> USB500 fallback.

Type-C Rp detection is hardware implemented in USB page. No PD negotiation or USB enumeration/suspend-controlled 100mA mode. Validate intended sources and full-load thermal behavior.

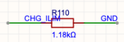
R111=1k，充电标称890mA；R112=69.8k，安全计时典型9.31h；DPPM优先供系统。

样机条件：使用4.2V充满的单节保护电池及贴附电芯的10k NTC；电流和温升待验证。



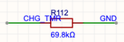


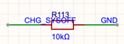








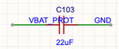


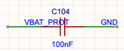



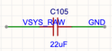


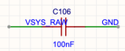


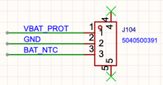


## 12_POWER_BUTTON / 长按电源键；取消 J111 双刀机械开关

开机：SW4 长按约 2.2 秒；关机：长按约 3.2 秒后通知主控保存，随后关闭供电。

U110 由 VSYS_RAW 常供电；EN 控制 TPS63802。R116 保留为 BUCK_EN 上拉。

MAIN_IN 并入 VSYS_RAW；BOOST_5V 并入 +5V_SYS。BQ24075 充电路径不变。

关机时 3V3 下降，BOOST_ENABLE 经 R120 下拉，TPS61023 进入输出断开状态。

GPIO14=PWR_INT_N；GPIO6=PWR_KILL_N，仅使用输入高阻或开漏；GPIO44/43 保留 UART0 RX/TX。

R141 从 3V3 上拉 KILL，启动无需固件在 512ms 内接管；复位和 USB 下载不会主动关电。

INT 有效后固件应在 700ms 内关闭负载并保存，再拉低 KILL；硬件约 0.8~1.3s 强制关闭。

关机保留充电和按键检测。BOOT / RESET 留板上调试，外壳仅露 POWER。

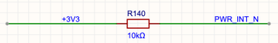
时序含电容/控制器误差；USB 插拔、低电池、故障关机、残压与实物待测。





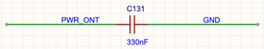


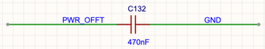









| | | | |
|---|---|---|---|
| 原理图 | 07_MAIN_LINK | 创建日期 | 2026-10-26 |
| | | 更新日期 | 2026-10-26 |
| 板子 | 02_电源板_Power | 图页 | 12_Main_switch |
| 绘制 | | Spirit_V080_Three_Board_Layout_20261026 | |
| 审阅 | | | |
| | | 版本 | 尺寸 |
| 嘉立创EDA | | V1.0 | A2 |
| | | 页 4 共 7 | |
| | | 嘉立创EDA | |

**SPIRIT / 13_Buck_3V3 / 原理图复核版 2026-10-18**

TPS63802 synchronized to schematic and PCB. V074K local placement repaired; board remains UNROUTED pending manual routing.

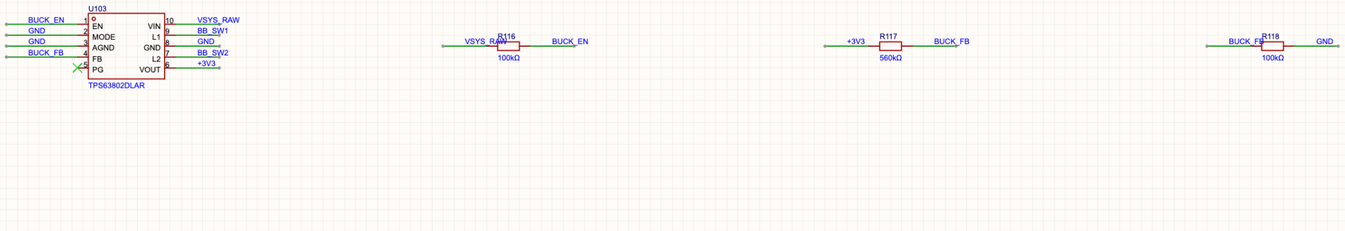
1A main-rail budget; protected 1S battery; low-battery firmware required. No full power-stage simulation/thermal release.

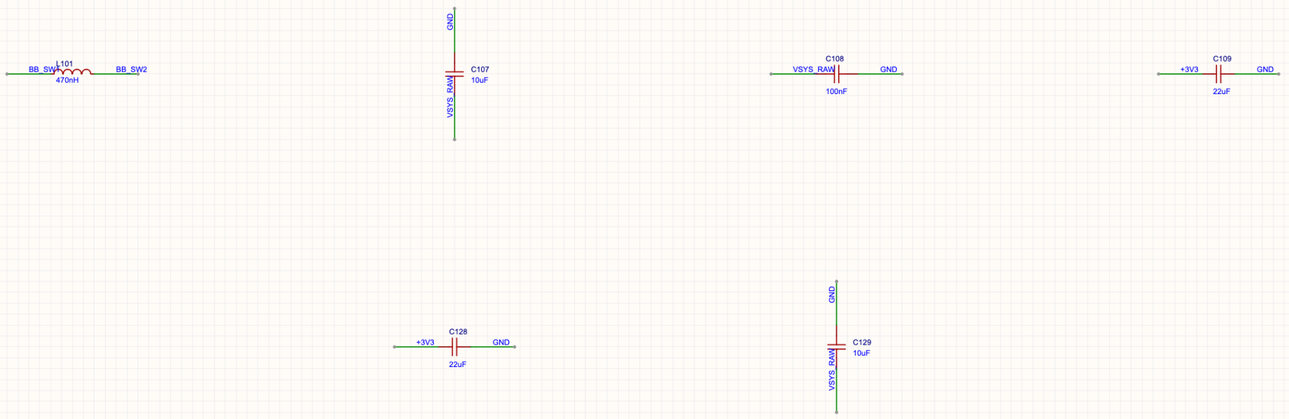
Input: 2x10uF + 100nF; output: 2x22uF. Verify effective Cin>=4uF and Cout>=7uF with bias/temp.

L101=0.47uH C408353 between L1/L2; do not reuse 2.2uH buck connection. MODE=GND, PG=NC.

R117=560k 0.1%, R118=100k 0.1%; VOUT=0.5*(1+560/100)=3.300V nominal. 2026-10-21.

## 13_MAIN_BUCK_BOOST / TPS63802DLAR C2845237 / 3.3V nominal

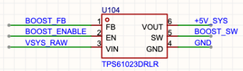


**SPIRIT / 14_Boost_5V / 原理图复核版 2026-10-18**

TPS61023输出直接为+5V_SYS，标称4.992V；J107和外部双刀开关已取消，由BOOST_ENABLE控制。

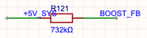
+5V_SYS supplies MAX98357A and U112 accessory limiter; J7 pin 1 is +5V_ACC. Screen logic/backlight use +3V3; actual panel FPC/backlight remain unconfirmed.









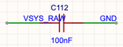


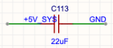













**SPIRIT / 15_Battery_measurement / 原理图复核版 2026-10-18**

